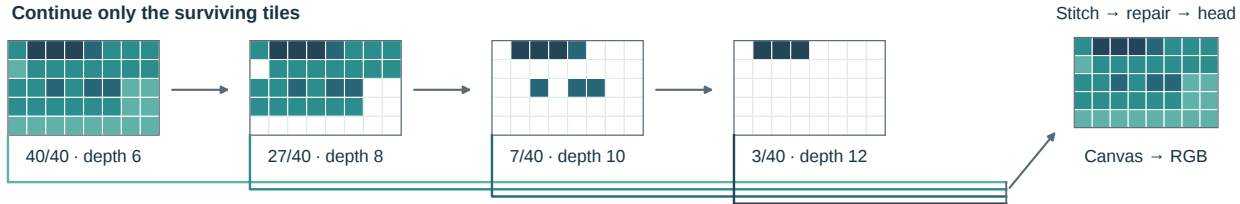


a Recover once; assign a stopping depth to each tile

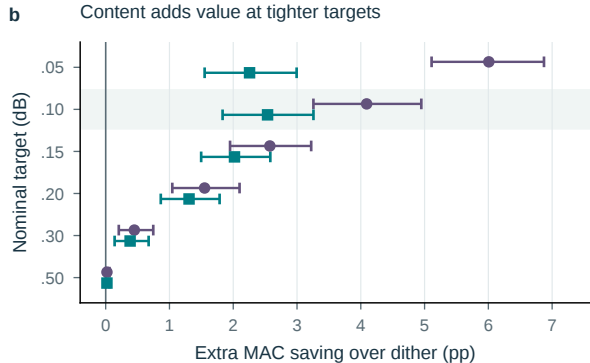
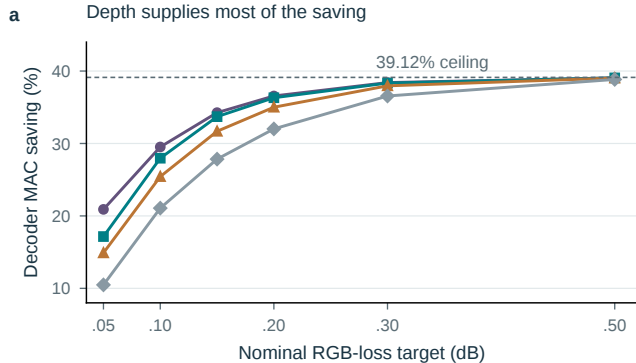


b Continue only the surviving tiles



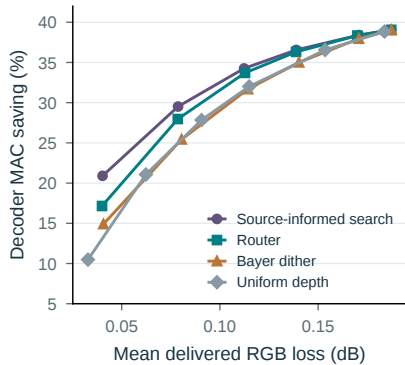
Pointwise exit adapters; deepest exit uses identity

Source-informed search Router Bayer dither Uniform depth

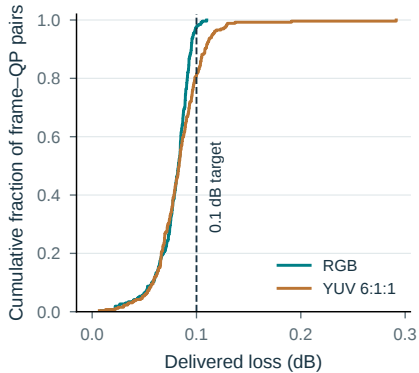


Paired 95% sequence intervals · source-calibrated controls · arithmetic savings, excluding routing and signalling

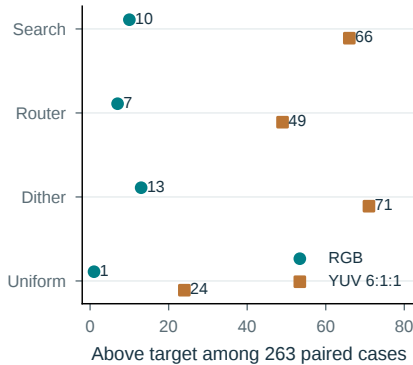
a Achieved quality differs across rules



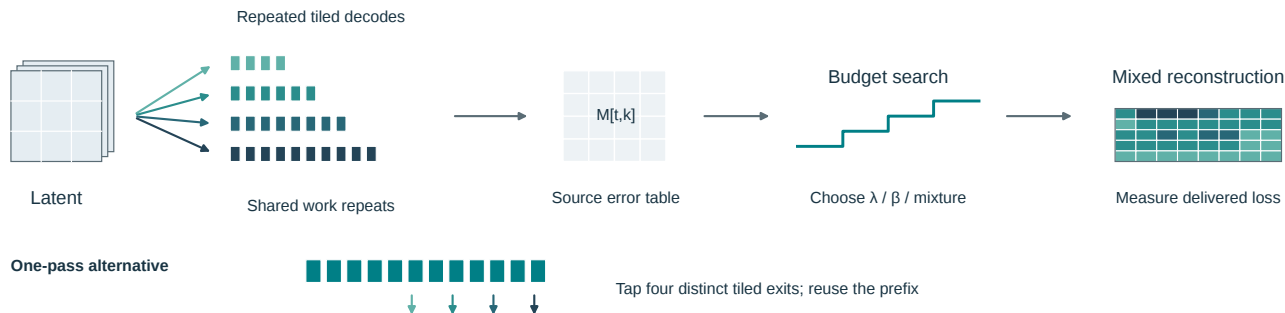
b Reporting differs from selection



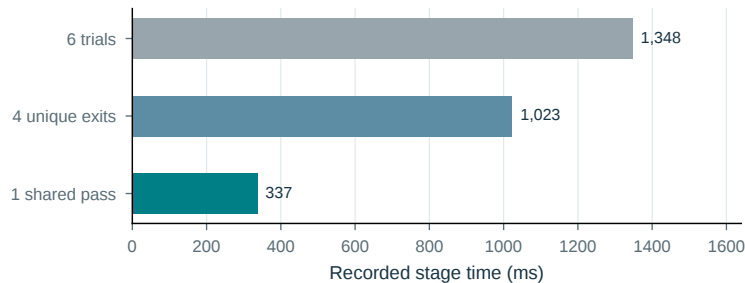
c Losses above the nominal target



a The source is available at the encoder, but the error table must be computed



b Recorded table construction time



Historical timing: 8 frames $\times$ 3 repetitions at QP32. No new end-to-end latency measurement.

c Calibration changes the cost

Fixed, held-out β

Router uses decoded features.
No per-frame source table is required.

Per-frame source-calibrated β

The plotted budget sweep uses $M[t, k]$.
A fast router does not remove that cost.

a One source, four allocations at the 0.1 dB target

Source luma



videoSRC05

QP32

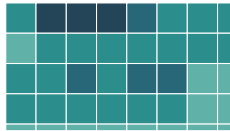
Search



28.8% MAC saved

0.085 dB loss

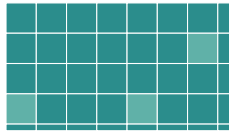
Router



26.1% MAC saved

0.085 dB loss

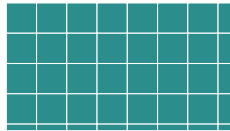
Bayer dither



25.3% MAC saved

0.092 dB loss

Uniform



24.2% MAC saved

0.082 dB loss



6



8



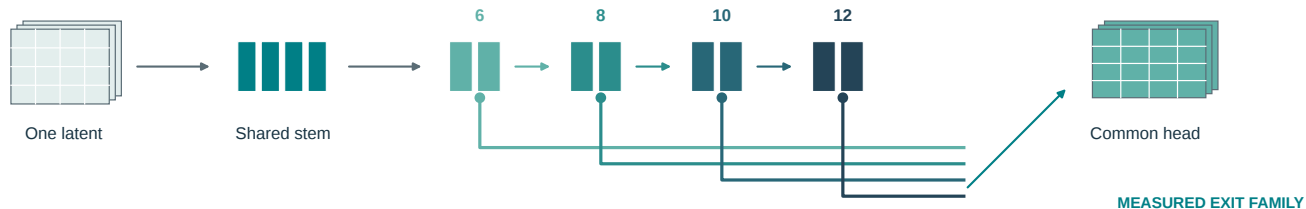
10



12

executed trunk blocks

a Shared early exit: one representation, nested synthesis



b Independent depth controls: each codec learns its own representation



Same-recipe D12 is an additional control. The released anchor alone does not isolate depth.

c Later routing study: select before encoding; signal which expert owns the payload



